

Math 301 Syllabus

Advanced Mathematics for Scientists and Engineers

Instructor:

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Office Hours:

By Appointment
Open Door Policy

Course Information:

Text: *Calculus, Early Transcendental Functions, (Fourth ed.)*, by Larson, Hostetler and Edwards
Prerequisites: C or better in MA 215 and MA 311
Department Head: LTC Troy Siemers

Grading:

Tests:	30% (3 Tests)
Homework:	40%
Final Exam:	30%

Homework:

Homework will consist of both hand-written and online assignments. Hand-written homework assignments should be organized and rigorous. No scratch-work solutions! The due dates for these assignments will be announced in class and/or posted on the class website. Online homework will be assigned for each topic that we cover and must be submitted by the due date listed on the homework website.

Online Homework Website: <https://euler.vmi.edu/webwork2>

Your class will be listed as “MA_301_S15”. Sign in using your VMI login credentials. [Note: Some browsers display security warnings. Disregard these warning and confirm access to this website.]

Course Topics:

Vector Analysis: Vector Fields, Line Integrals, Conservative Vector Fields, Green’s Theorem, Parameterization of Lines and Surfaces, Divergence Theorem, Stokes’ Theorem
Miscellaneous Topics: Fourier Series, Complex Variables, Partial Differential Equations

Attendance Policy:

The following is the attendance policy at VMI. Academic excellence is best achieved through consistent class attendance. The maximum allowed percentage of class absences is 30%. No categories of absences (academic, athletic, guard, 3.0 cuts, etc.) will be exempt from that percentage. When a cadet reaches 20%

absences, the instructor issues a written warning and the cadet must sign a receipt that he/she is aware of the absences incurred. Upon reaching 30% absences the cadet is referred to the Dean for appropriate action. Normally a cadet who exceeds the 30% absences will be required to withdraw from the course with a W or a WF.

Students with Disabilities:

VMI abides by Section 504 of the Rehabilitation Act of 1973 and the Americans with Disabilities Act of 1990, which mandate reasonable accommodations be provided for cadets with documented disabilities. If you have a registered disability and may require some type of instructional and/or examination accommodations, please contact me early in the semester so that I can provide or facilitate provision of accommodations you may need. If you have not already done so, you will need to register with the Office of Disabilities Services, the designated office on campus to provide services for cadets with disabilities. The office is located on the 2nd floor of the VMI Health Center. Please call or stop by the office of LTC Jones, Director of Disabilities Services, for more information, 464-7667 or jonesl10@vmi.edu.

Your Responsibility:

Make sure that you understand all of the policies in this document, especially the policies concerning work for grade. If you do not understand or if you disagree with any part of this document you must contact the instructor immediately.

Work for Grade:

"Work for grade" is defined as any work presented to an instructor for a formal grade or undertaken in satisfaction of a requirement for successful completion of a course or degree requirement. All work submitted for grade is considered the cadet's own work. **"Cadet's own work"** means that he or she has composed the work from his or her general accumulation of knowledge and skill except as clearly and fully documented and that it has been composed especially for the current assignment. No work previously submitted in any course at VMI or elsewhere will be resubmitted or reformatted for submission in a current course without the specific approval of the instructor.

In all work for grade, failure to distinguish between the cadet's own work and ideas and the work and ideas of others is known as **plagiarism**. Proper documentation clearly and fully identifies the sources of all borrowed ideas, quotations, or other assistance. The cadet is referred to the VMI authorized handbook for rules concerning quotations, paraphrases, and documentation.

In all written work for grade, the cadet must include the words **"HELP RECEIVED"** conspicuously on the document, and he or she must then do one of two things: (1) state "none," meaning that no help was received except as documented in the work; or (2) explain in detail the nature of the help received. In oral work for grade, the cadet must make the same declaration before beginning the presentation. Admission of help received may result in a lower grade but will not result in prosecution for an honor violation.

Cadets are prohibited from discussing the contents of a quiz/exam until it is returned to them or final course grades are posted. This enjoinder does not imply that any inadvertent expression or behavior that might indicate one's feeling about the test should be considered a breach of honor. The real issue is whether cadets

received information, not available to everyone else in the class, which would give them an unfair advantage. If a cadet inadvertently gives or receives information, the incident must be reported to the professor and the Honor Court. Each cadet bears the responsibility for familiarizing himself or herself thoroughly with the policies stated in this section, with any supplementary statement regarding work for grade expressed by the academic department in which he or she is taking a course, and with any special conditions provided in writing by the professor for a given assignment. If there is any doubt or uncertainty about the correct interpretation of a policy, the cadet should consult the instructor of the course. There should be no confusion, however, on the basic principle that it is never acceptable to submit someone else's work, written or otherwise, formally graded or not, as one's own.

The violation by a cadet of any of these policies will, if he or she is found guilty by the Honor Court, result in his or her being dismissed from VMI. Neither ignorance nor professed confusion about the correct interpretation of these policies is an excuse.

Departmental Work for Grade Policy:

The faculty of the Department of Applied Mathematics fully supports the Institute Work for Grade Policy as passed by the Academic Board and approved by the Superintendent. Because of this, every faculty member in our department will promptly contact any cadet who has submitted work for grade without the expression "**HELP RECEIVED**" followed by the word "none" or an explanation of the nature of the help received. The faculty member will inform the cadet that the work will not be graded until the cadet adds the required "**HELP RECEIVED**" statement to the work.

Writing Center:

The use of the Writing Center is approved for all courses offered by the Department.

Mathematics Education and Resource Center (MERC):

The use of the MERC and its Open Math Lab (OML) is approved for all courses offered by the Department, unless otherwise specified by your professor or instructor.

Tutoring, Peer Collaboration, Academic Center:

It is our department's policy that any work submitted for grade precludes the aid of any tutors or peer collaboration unless specifically stated otherwise by the instructor in the course syllabus. Tutors and peer collaboration are authorized for assigned work that is not graded, such as homework or drill exercises.

Use of Computer Aids:

The use of software packages is **not** permitted unless specifically stated otherwise by the instructor in the course syllabus. However, spelling, style, and grammar, checkers are approved for cadet use.

The use of a calculator is to be determined by the instructor as appropriate for each course.